Toward a Type-Theoretic Framework for Linear Mode Connectivity: Univalence and Path-Finding in Weight Spaces

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Abstract

Permutation alignment gives a standard group-action explanation for many observations in Linear Mode Connectivity (LMC): two trained networks may be far apart as raw parameter vectors while close after choosing representatives in the same symmetry quotient. This paper asks what extra structure a Homotopy Type Theory (HoTT) view can add to that classical picture. We present a dictionary in which architectures are types, trained weights are terms, permutation quotients are Higher Inductive Types, and alignment certificates are proof objects that can be composed and transported. The resulting univalent interpretation is deliberately conditional: it organizes discrete symmetry evidence as identity paths, but a separate realization functor and Euclidean smoothness assumptions are needed before one obtains a low-loss curve in parameter space. Our experiments are calibration checks rather than new empirical claims: MNIST MLPs reproduce the familiar $> 98\%$ barrier reduction after permutation alignment, while CIFAR-10 ResNet20 runs illustrate that BatchNorm statistics require REPAIR-style renormalization in our interpolation protocol.

1. Introduction

Linear Mode Connectivity (LMC) studies when two trained parameter vectors $\theta_A, \theta_B \in \mathbb{R}^D$ can be joined by a path of low empirical risk (Frankle et al., 2020; Garipov et al., 2018). For the linear segment $\theta_\alpha = (1-\alpha)\theta_A + \alpha\theta_B$, we use the standard barrier

$$\Delta(\theta_A, \theta_B) = \max_{\alpha \in [0,1]} L(\theta_\alpha) - \tfrac{1}{2}\big(L(\theta_A) + L(\theta_B)\big), \quad (1)$$

where L is the empirical loss. A large Δ often disappears after permuting hidden units of one network, as in Git-Re-Basin (Ainsworth et al., 2023) and related permutation-alignment methods (Entezari et al., 2022). In classical terms, an architecture has a group G of function-preserving parameter symmetries; alignment searches for $\pi \in G$ so that θ_A and $\sigma_\pi(\theta_B)$ are close representatives of their quotient classes.

The standard language of group actions and quotient spaces is sufficient to state this empirical mechanism. Our aim is not to replace it, nor to claim that HoTT alone proves new Euclidean LMC theorems. Instead, we ask whether HoTT supplies useful formal structure for the *certificates* produced by alignment algorithms. In a type-theoretic reading, a type is a structured space, a term is a point of that space, an identity type is the type of paths between two terms, and an equivalence is an invertible structure-preserving map. Higher Inductive Types (HITs) allow quotients to be specified by adding paths as constructors rather than by selecting canonical representatives. The Univalence Axiom then states, at the level of types in a universe, that equivalences can be treated as identities (Program, 2013; Grayson, 2018).

This distinction is the main conceptual contribution. Group actions tell us that changing representatives within one symmetry orbit preserves the realized function. HoTT gives a proof-relevant account of that statement: a permutation is not only an element of G, but a quotient-path witness for replacing B by $\sigma_\pi(B)$. Separately, the alignment objective gives metric evidence that $\sigma_\pi(B)$ is close to A. A low-loss interpolation claim needs both pieces: formal orbit evidence and Euclidean regularity. To turn identity/path evidence into an actual curve $[0, 1] \to \mathbb{R}^D$, one must construct a realization map and assume regularity of L along the realized curve. We therefore present the univalent LMC statement as a conditioned conjectural interface between discrete alignment and continuous optimization.

Our contributions are: (i) a self-contained dictionary between permutation alignment and HoTT notions, calibrated

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for readers who know LMC but not type theory; (ii) a precise separation between orbit/path evidence, endpoint-distance evidence, and Euclidean loss bounds; (iii) calibration experiments reproducing known MNIST and CIFAR-10 phenomena under this vocabulary; and (iv) appendix material that records the elementary Euclidean facts and marks the cohesive-realization step as an open formalization target.

2. Related Work

Linear Mode Connectivity. Early investigations revealed that while independent training runs produce models in disjoint basins, these minima are often connected by continuous, non-linear paths of low loss (Garipov et al., 2018; Draxler et al., 2019). Frankle et al. (Frankle et al., 2020) demonstrated that LMC holds empirically even under simple linear interpolation when networks are initialized from the same pre-trained weights.

Permutation Alignment and Model Merging. High barriers can be artifacts of permutation invariance (Brea et al., 2019). Git-Re-Basin (Ainsworth et al., 2023) explicitly matches hidden units before interpolation, while model soups and task arithmetic study averaging and editing in broader model-merging settings (Wortsman et al., 2022; Ilharco et al., 2023). Pure permutation alignment is often insufficient in architectures with normalization layers: REPAIR (Jordan et al., 2023) recalibrates activations to stabilize interpolation. Other approaches optimize geometric paths directly, including Fisher–Rao/JSD geodesics (Tan et al., 2023), layerwise paths (Tian et al., 2025), and optimal-transport-inspired fusion methods (Imfeld et al., 2024).

Formal LMC Guarantees. Ferbach et al. (Ferbach et al., 2024) prove that sufficiently wide two-layer networks trained with SGD are linearly connected with high probability, using optimal transport and Wasserstein convergence rates to bound width. Zhao et al. (Zhao et al., 2025) construct explicit curves for minima related by continuous symmetries, deriving curvature conditions under which approximate LMC holds. These works provide Euclidean guarantees. Our framework is complementary: it packages alignment evidence as identity/path data, while leaving the Euclidean realization and loss control to additional assumptions.

Homotopy Type Theory in ML. While HoTT (Program, 2013) and Univalent Foundations (Grayson, 2018) have revolutionized formal verification and synthetic topology, their application to deep learning remains nascent. To our knowledge, this is the first explicit framing of LMC via Univalence.

3. Formalizing Weight-Space as Types

We now make the dictionary explicit. In classical ML, a neural network is a parameterized function $f_\theta : \mathcal{X} \to \mathcal{Y}$ with $\theta \in \mathbb{R}^D$. In the type-theoretic presentation, the architecture is a type $\mathcal{N}$, a trained parameter vector is a term $A : \mathcal{N}$, and a symmetry certificate is a path or equivalence witness involving such terms.

3.1. Neural Networks as Cartesian Products

A feedforward layer l computes $h^{(l)} = \sigma(W^{(l)} h^{(l-1)} + b^{(l)})$, where the space of weights and biases for a given layer dimension forms a fixed Type $\mathcal{L}$. A complete L-layer architecture is a Cartesian product:

$$\mathcal{N} \equiv \mathcal{L}_1 \times \mathcal{L}_2 \times \cdots \times \mathcal{L}_L \tag{2}$$

An instantiated network is a term $A : \mathcal{N}$. For an MLP with hidden widths $n_1, \ldots, n_{L-1}$, the permutation group is $G = \prod_{l=1}^{L-1} \mathfrak{S}_{n_l}$. Each $\pi \in G$ induces a linear parameter action $\sigma_\pi : \mathcal{N} \to \mathcal{N}$ that permutes hidden units and applies inverse permutations in adjacent layers. This action preserves the realized function, $f_{\sigma_\pi(\theta)} = f_\theta$ (proof in Section A). The quotient $\mathcal{N}/G$ forgets which representative of a symmetry orbit was chosen.

Definition 3.1 (Types as Orbits via HITs)**.** The *quotient weight type* $\mathcal{N}//G$ is constructed as a HIT. Its point constructors include standard models $A : \mathcal{N}$, and its path constructors explicitly add identity paths for the group action:

$$\mathsf{glue}(A, \pi) : [A] =_{\mathcal{N}//G} [\sigma_\pi(A)].$$

Thus the quotient is not merely a set of equivalence classes; it stores the evidence that two representatives are connected by a particular symmetry.

3.2. Permutation Symmetry as Isomorphism

Since permuting neurons with the corresponding inverse permutation of the next layer leaves output invariant (Hecht-Nielsen, 1990), any $\pi \in G$ induces an equivalence $\varphi_\pi : \mathcal{N} \xrightarrow{\sim} \mathcal{N}$ (functional invariance and the generic orbit size $|G| = \prod_l n_l!$ are recorded in Section A).

If model B can be permuted to align with model A, the alignment algorithm returns a permutation π and two distinct pieces of evidence. The first is formal orbit evidence, supplied by the quotient constructor,

$$p_{\text{orbit}}(\pi) : [B] =_{\mathcal{N}//G} [\sigma_\pi(B)]. \tag{3}$$

The second is metric evidence that the chosen representative is close to A. For a single matched layer or block, the usual objective is

$$\pi^* = \arg\max_{\pi \in \mathfrak{S}_n} \mathrm{Tr}(W_A^\top P_\pi W_B), \tag{4}$$

equivalently minimizing $\|W_A - P_\pi W_B\|_F^2$. This is solved by the Linear Sum Assignment Problem (LSAP) (Kuhn, 1955). Full Git-Re-Basin applies such assignments across layers; in our formal statements, $\varepsilon = \|\theta_A - \sigma_{\pi^*}(\theta_B)\|$ denotes the post-alignment endpoint distance. The small value of ε, not the quotient path alone, is what enters the Euclidean barrier bound.

4. The Univalence Axiom for Model Merging

The critical type-theoretic bridge is the **Univalence Axiom** (Grayson, 2018). It should be read here as a way to organize symmetry evidence, not as a stand-alone theorem about neural loss landscapes.

4.1. Conceptualizing Univalence in ML

In standard set theory, isomorphism and equality are distinct. HoTT introduces the identity type $\mathcal{A} =_U \mathcal{B}$, interpreted as the space of paths between types in a universe U. Univalence says that the canonical map from paths to equivalences is itself an equivalence:

$$(\mathcal{A} =_U \mathcal{B}) \simeq (\mathcal{A} \simeq \mathcal{B}). \tag{5}$$

For network *terms* $A, B : \mathcal{N}$, the more direct construction is the HIT quotient path $\mathsf{glue}(A, \pi) : [A] = [\sigma_\pi(A)]$. Univalence enters one level higher: if we package symmetry orbits, layers, or architectures as types, then equivalences between those packages can be treated as paths in the formal universe. A cohesive or spatial semantic model would be needed to interpret such formal paths as continuous geometric curves.

4.2. Path-Finding via Univalence

When an alignment routine yields π, the quotient HIT records $p_{\mathrm{orbit}}(\pi) : [B] = [\sigma_\pi(B)]$. If higher-level orbit packages $\mathcal{O}_A$ and $\mathcal{O}_B$ include both this formal representative-change witness and the metric evidence needed for interpolation, an equivalence between such packages can be converted by univalence into an abstract universe-level path:

$$\mathrm{ua}(e) : \mathcal{O}_A =_U \mathcal{O}_B. \tag{6}$$

This is the proof-search interpretation of alignment: the algorithm constructs part of the data that inhabits an identity/equivalence interface in the quotient presentation. The statement becomes a claim about LMC only after a realization functor maps that evidence to a curve in $\mathbb{R}^D$.

Quantifying the realized path requires a map $[\![\cdot]\!]$ from type-theoretic paths to continuous curves $[0, 1] \to \mathbb{R}^D$. Under standard second-order assumptions, if both endpoints are stationary and the Hessian along the aligned segment is bounded above by λI, then the linear-interpolation barrier is at most $\frac{\lambda}{8}\|\sigma_{\pi^*}(\theta_B) - \theta_A\|^2$ (Section A.2). Alignment therefore reduces a concrete upper-bound proxy by reducing the endpoint distance. Low-loss connectivity requires both ingredients: structural equivalence evidence and Euclidean regularity.

4.3. What the HoTT Layer Adds

The practical alignment algorithm remains an assignment problem. The HoTT layer adds three pieces of structure useful for later formal work. First, paths are proof-relevant: two different permutations can be different witnesses even when they connect the same quotient classes. Second, paths compose, so layerwise, blockwise, or taskwise merges can be represented as explicit proof terms rather than as informal bookkeeping. Third, transport along paths gives a natural place to state which properties should be preserved by merging, such as functional equality, bounded loss, or calibrated activation statistics. These advantages are organizational and verification-oriented; they do not remove the need for the Euclidean bounds above.

Concretely, the object we would like a future verification system to check is not merely "models A and B merge." It is a certificate consisting of a representative change π, a quotient witness $p_{\mathrm{orbit}}(\pi)$, a numerical endpoint-distance bound ε, and any architecture-specific side conditions such as BatchNorm recalibration. The appendix formalizes this certificate and proves the resulting conditional linear-barrier bound. This keeps the role of HoTT narrow: it structures the symbolic part of the certificate, while analysis controls the loss along the realized curve.

5. Empirical Proof-of-Concept

We use experiments only to calibrate the formal vocabulary against known LMC phenomena. The goal is not to introduce a new alignment method or benchmark, but to show where the type-theoretic interpretation matches and where additional geometric or statistical assumptions enter.

5.1. Experimental Setup

We define our Type $\mathcal{N}$ as a three-layer MLP with hidden dimension 512, trained on the MNIST dataset (Lecun et al., 1998). We train two instantiations, Model A and Model B, using different random seeds. Both converge to equivalent test accuracies but reside in different regions of the parameter space. We implement a simplified Git-Re-Basin (Ainsworth et al., 2023), computing optimal equivalences via the Linear Sum Assignment solver across 3 independent random seed pairs.

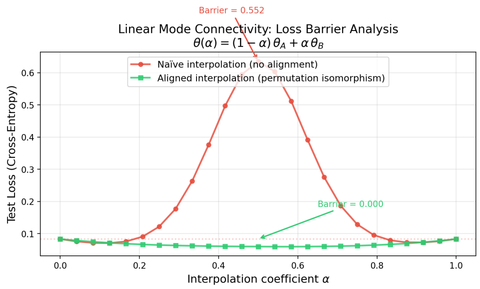

Figure 1. LMC on MNIST. Naive interpolation encounters a large loss barrier, while permutation alignment flattens the barrier. In our framework, the permutation is the quotient-path witness; the low-loss Euclidean segment additionally relies on smoothness of the realized loss landscape.

5.2. Results and Discussion

Naive interpolation yields a large loss barrier; after choosing π^* by weight alignment, the barrier collapses (Table 1, $>$ 98% reduction). This reproduces the standard permutation-alignment phenomenon and supplies a concrete example of the quotient-path witness $p_{\text{orbit}}(\pi^*)$ together with small endpoint distance.

Table 1. Summary of LMC Barrier Reduction (3 trials, width 512)

Interpolation Method	Peak Loss Barrier	Min Accuracy
Naive (No Alignment)	0.459 ± 0.025	92.1%
Aligned (Permutation Isom.)	$\mathbf{0.006 \pm 0.002}$	$\mathbf{98.0\%}$

5.3. Type Dimension and Isomorphism Density

We evaluate the loss barrier across hidden dimensions 32–1024 to check how the post-alignment distance proxy changes with the size of the permutation orbit.

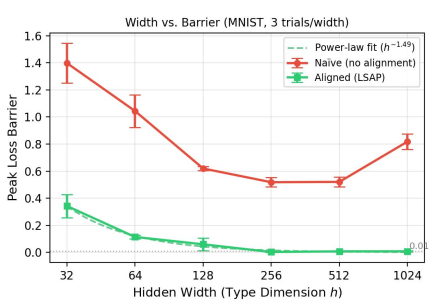

Figure 2. Loss barrier vs. hidden width (MNIST, 3 trials/width). The aligned barrier decays rapidly (power-law $h^{-1.49}$) and falls below 0.01 at width ≥ 256; the naïve barrier remains large throughout.

Figure 2 shows that, on MNIST, the aligned barrier decays rapidly with width (power-law $h^{-1.49}$, better fit than exponential), falling below 0.01 at width ≥ 256. This is consistent with the distance-bound proxy in Corollary A.6: a larger orbit $|G| = h!^2$ gives the assignment procedure a richer search space, reducing the observed endpoint mismatch ε.

5.4. Scaling to ResNets and Activation Renormalization

We also evaluate ResNet20 on CIFAR-10 (Krizhevsky, 2009), where Batch Normalization can cause feature-statistic mismatch that defeats pure weight interpolation.

In our ResNet20 experiments (Figure 3), we evaluate Activation Renormalization (REPAIR (Jordan et al., 2023)) and purely geometric baselines (LLPF (Tian et al., 2025), spherical geodesics (Tan et al., 2023)). REPAIR is the primary mechanism in this protocol: without it, loss barriers exceed 10 nats regardless of alignment; with it, barriers fall below 0.75. Alignment provides a smaller additional benefit. The takeaway for the formal story is that a quotient-path witness is not enough: transported properties may include activation-statistic calibration, not just functional equivalence modulo permutation.

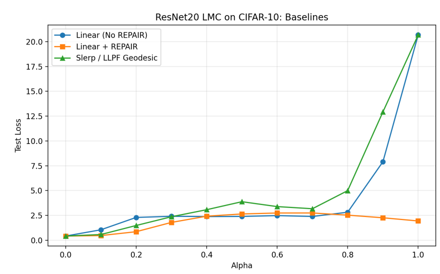

Figure 3. ResNet20 on CIFAR-10. REPAIR is the primary mechanism for barrier reduction in our protocol; geometric geodesics (LLPF, slerp) fail without correcting BN statistic mismatch. Alignment provides a further small reduction.

5.5. Experimental Methodology and Limitations

For the MNIST MLP experiments, both models were trained with Adam (lr $= 10^{-3}$, batch size 256) for 10 epochs; barriers use Equation (1) over 25 uniformly spaced interpolation points, with confidence intervals across 3 seed pairs. For ResNet20, models were trained with SGD (momentum 0.9, weight decay 10^{-4}, cosine schedule) for 200 epochs; BN running statistics are recalibrated independently at each interpolation point α by a forward pass over 2,000 training samples with the interpolated weights frozen.

These experiments are intentionally limited. They reproduce established effects with small trial counts, and they do not validate the univalent conjecture by themselves. Their role is to identify the empirical objects that a formal account must track: quotient witnesses, endpoint distances, Hessian/smoothness assumptions, and normalization statistics.

6. Conclusion

We presented a type-theoretic account of permutation alignment for LMC. The key point is modest but precise: group actions explain the symmetry quotient, while HoTT turns quotient membership and alignment into composable identity evidence. Low-loss Euclidean interpolation is a further geometric claim, requiring a realization map and smoothness assumptions. Future work should construct such a cohesive semantics for neural parameter spaces and connect it to proof assistants capable of checking model-merging certificates.

Software, Data, and Accessibility

Code and reproducibility scripts are available at `https://github.com/suayptalha/univalent-lmc`. Figures include screen-reader-compatible captions.

Impact Statement

This paper develops a theoretical framework connecting Homotopy Type Theory to the geometry of neural network weight spaces. The primary contributions are foundational: we provide a formal language for reasoning about model merging and linear mode connectivity that may inform future work in distributed training, model reuse, and continual learning. By treating symmetry changes as explicit proof objects, the framework points toward checkable model-merging certificates. Such certificates could help identify failure modes and boundary conditions that purely empirical approaches may miss.

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A. Euclidean Grounding: Proofs

We give proofs for the elementary Euclidean facts used in the main text. These results are standard consequences of permutation invariance and second-order smoothness; they are included to fix notation and to separate established geometry from the conjectural type-theoretic realization step.

A.1. Group Action and Functional Invariance

For a two-hidden-layer MLP $\theta = (W_1, b_1, W_2, b_2, W_3, b_3)$ with hidden dimension h, let $G = \mathfrak{S}_h \times \mathfrak{S}_h$. Define

$$\sigma_{(\pi_1,\pi_2)}(\theta) := (P_{\pi_1} W_1,\ P_{\pi_1} b_1,\ P_{\pi_2} W_2 P_{\pi_1}^{-1},\ P_{\pi_2} b_2,\ W_3 P_{\pi_2}^{-1},\ b_3).$$

Proposition A.1 (Group Action and Functional Invariance). *$(\pi_1, \pi_2) \mapsto \sigma_{(\pi_1,\pi_2)}$ is a group action of G on $\mathcal{N}$, and $f_{\sigma_{(\pi_1,\pi_2)}(\theta)}(x) = f_\theta(x)$ for all x.*

Proof. Group action. Identity is immediate ($P_{\text{id}} = I$). Compatibility follows from $P_\pi P_\rho = P_{\pi\circ\rho}$, since $(P_\pi P_\rho)_{ij} = \sum_k \mathbf{1}[\pi(k){=}i]\mathbf{1}[\rho(j){=}k] = \mathbf{1}[\pi(\rho(j)){=}i] = (P_{\pi\circ\rho})_{ij}$.

Functional invariance. Let $h_1 = \sigma(W_1 x + b_1)$ be the first hidden activation of θ. Under the transformation, $h_1' = \sigma(P_{\pi_1} W_1 x + P_{\pi_1} b_1) = P_{\pi_1}\sigma(W_1 x + b_1) = P_{\pi_1} h_1$, where we used that ReLU is elementwise, so it commutes with any permutation matrix. Similarly $h_2' = \sigma(P_{\pi_2} W_2 P_{\pi_1}^{-1} \cdot P_{\pi_1} h_1 + P_{\pi_2} b_2) = P_{\pi_2}\sigma(W_2 h_1 + b_2) = P_{\pi_2} h_2$. The output is $W_3 P_{\pi_2}^{-1} \cdot P_{\pi_2} h_2 + b_3 = W_3 h_2 + b_3 = f_\theta(x)$. □

Corollary A.2 (Orbit Equivalence and Invertibility). *Each orbit $[\theta] = \{\sigma_\pi(\theta) : \pi \in G\}$ consists entirely of parameters representing the same function. Each $\sigma_{(\pi_1,\pi_2)}$ is a bijection with inverse $\sigma_{(\pi_1^{-1},\pi_2^{-1})}$. For generic networks (no two neurons in the same layer with identical weight rows), the stabiliser is trivial and $|[\theta]| = |G| = h!^2$.*

A.2. Barrier Invariance and Loss Bounds

Lemma A.3 (LSAP Minimises Alignment Distance). *The LSAP objective $\pi^* = \arg\max_\pi \operatorname{Tr}(W_A^\top P_\pi W_B)$ is identical to $\arg\min_\pi \|P_\pi W_B - W_A\|_F^2$.*

Proof. Expand $\|P_\pi W_B - W_A\|_F^2 = \|W_A\|_F^2 + \|W_B\|_F^2 - 2\operatorname{Tr}(W_A^\top P_\pi W_B)$ using $\|P_\pi W_B\|_F^2 = \|W_B\|_F^2$ (since P_π is orthogonal). The first two terms are constant over π, so minimising the squared norm is equivalent to maximising the trace. □

Proposition A.4 (Barrier Invariance Under Simultaneous Orbit Shift). *Let $B(\theta_A, \theta_B) := \max_{\alpha\in[0,1]} \mathcal{L}((1-\alpha)\theta_A + \alpha\theta_B) - \frac{1}{2}(\mathcal{L}(\theta_A) + \mathcal{L}(\theta_B))$. For any $\pi \in G$, $B(\sigma_\pi(\theta_A), \sigma_\pi(\theta_B)) = B(\theta_A, \theta_B)$. Permuting only θ_B changes the barrier; this is what makes alignment non-trivial.*

Proof. The action is linear in weights, so $(1-\alpha)\sigma_\pi(\theta_A) + \alpha\sigma_\pi(\theta_B) = \sigma_\pi((1-\alpha)\theta_A + \alpha\theta_B)$. By Corollary A.2 the loss is orbit-invariant, so the loss along the interpolation and thus the barrier are unchanged. □

Proposition A.5 (Symmetric Barrier Bound). *Let $\mathcal{L}$ be twice differentiable with $\nabla^2\mathcal{L} \preceq \lambda I$ throughout the segment from θ_A to $\pi(\theta_B)$. If $\nabla\mathcal{L}(\theta_A) = \nabla\mathcal{L}(\pi(\theta_B)) = 0$ and $\varepsilon = \|\pi(\theta_B) - \theta_A\|$, then*

$$\max_{t\in[0,1]} \mathcal{L}(\theta(t)) \le \max\big(\mathcal{L}(\theta_A), \mathcal{L}(\pi(\theta_B))\big) + \tfrac{\lambda}{8}\varepsilon^2,$$

where $\theta(t) = (1-t)\theta_A + t\,\pi(\theta_B)$.

Proof. Let $\Delta = \pi(\theta_B) - \theta_A$. Taylor expansion around θ_A (using $\nabla\mathcal{L}(\theta_A) = 0$):

$$\mathcal{L}(\theta(t)) \le \mathcal{L}(\theta_A) + \tfrac{t^2\lambda}{2}\|\Delta\|^2 \le \mathcal{L}(\theta_A) + \tfrac{t^2\lambda}{2}\varepsilon^2.$$

Taylor expansion around $\pi(\theta_B)$ (using $\nabla\mathcal{L}(\pi(\theta_B)) = 0$):

$$\mathcal{L}(\theta(t)) \le \mathcal{L}(\pi(\theta_B)) + \tfrac{(1-t)^2\lambda}{2}\varepsilon^2.$$

Taking the pointwise minimum of both bounds, the worst case is at $t = \frac{1}{2}$ (where they are equal), giving the $\frac{\lambda}{8}\varepsilon^2$ bound. □

Corollary A.6 (LSAP Minimises the Barrier Bound)**.** *Under the conditions of Proposition A.5, if the permutation objective is exactly the full endpoint distance $\|\sigma_\pi(\theta_B) - \theta_A\|$, then the minimizing π^* also minimizes the proxy $\frac{\lambda}{8}\varepsilon^2$ appearing in the barrier bound. In practical Git-Re-Basin, the global matching is often optimized through layerwise LSAP subproblems, so Lemma A.3 applies directly to each matched block and serves as a surrogate for the full endpoint-distance objective.*

B. Type-Theoretic Formalization Sketch

This appendix sketches the Martin-Löf Type Theory (MLTT) and HoTT objects used in the main text. It is not a completed formal proof of LMC. The open ingredient is a cohesive realization semantics that turns formal identity/path evidence into Euclidean curves with loss control.

B.1. MLTT Inference Rules for Weight Spaces

We define the universe of neural network weights U_{NN}. A specific layer dimension vector $\vec{d} = (d_{in}, d_{out})$ forms a base type.

Formation Rule for Weights: The parameters of a linear layer are defined as a matrix type, inherently a mapping between Euclidean spaces.

$$\frac{\Gamma \vdash d_{in} : \mathbb{N} \quad \Gamma \vdash d_{out} : \mathbb{N}}{\Gamma \vdash \mathcal{L}(d_{in}, d_{out}) : U_{NN}}$$

Identity Type (Id): In MLTT, equality is a type. If W_A and W_B are two weight configurations of type $\mathcal{L}$, the identity type states they are the same:

$$\frac{\Gamma \vdash W_A : \mathcal{L} \quad \Gamma \vdash W_B : \mathcal{L}}{\Gamma \vdash W_A =_{\mathcal{L}} W_B : U_{NN}}$$

The canonical constructor for the identity type is reflexivity (refl):

$$\frac{\Gamma \vdash W : \mathcal{L}}{\Gamma \vdash \mathsf{refl}_W : W =_{\mathcal{L}} W}$$

B.2. Path Induction and Loss Landscapes

In HoTT, the identity type $x = y$ is interpreted as the space of paths from x to y. Path induction states that to prove a property holds for all paths $p : x = y$, it suffices to prove it for the trivial path $\mathsf{refl}_x : x = x$.

Let Loss : $\mathcal{N} \to \mathbb{R}$ be the empirical risk. Path induction (J-elimination) allows *transport*: if a property P holds at A, and $p : A =_{\mathcal{N}} B$ is an identity path, then P can be transported to B. Applied to the property "loss $\leq c$", this is relevant only if the abstract path p is interpreted as a concrete curve in $\mathbb{R}^D$ that stays within the level set $\{L \leq c\}$—precisely the role of the realization functor discussed below.

B.3. Permutation Isomorphism Derivation

For a single matched block, finding the permutation used to change representatives reduces to minimizing the Frobenius distance between weight matrices:

$$\pi^* = \arg \min_{\pi \in \mathfrak{S}_n} \|W_A - P_\pi W_B\|_F^2 \tag{7}$$

Expanding the squared norm and dropping constant terms yields the equivalent trace maximization:

$$\pi^* = \arg \max_{\pi \in \mathfrak{S}_n} \mathrm{Tr}(W_A^\top P_\pi W_B) \tag{8}$$

which is resolved by the Linear Sum Assignment Problem (LSAP) (Kuhn, 1955).

B.4. Univalent Lifting: Commutative Diagram

Applying the Univalence Axiom to an equivalence $e : \mathcal{O}_A \simeq \mathcal{O}_B$ between suitably packaged orbit types gives an identity path $\mathrm{ua}(e) : \mathcal{O}_A =_U \mathcal{O}_B$. The diagram emphasizes that this is a formal path; the Euclidean curve requires a separate realization map:

$$\begin{tikzcd}
\mathcal{O}_A \arrow[rr, "e", "\simeq"'] \arrow[dr, "\mathrm{ua}(e)"'] & & \mathcal{O}_B \\
& \mathrm{Path}_U(\mathcal{O}_A, \mathcal{O}_B) \arrow[ur, "\text{transport}"'] &
\end{tikzcd}$$

Figure 4. Univalent lifting of an orbit-package equivalence to a formal identity path. A concrete low-loss curve requires an additional realization semantics.

B.5. Conditional Interface and Univalent LMC Conjecture

The Euclidean grounding is provided by Section A, which establishes concrete bounds without invoking cohesive semantics. The statement below is intentionally phrased as a conditional interface rather than as a new lemma: if a realization sends the formal path to the linear segment, then the second-order bound already proved in Proposition A.5 applies.

Remark B.1 (Conditional Euclidean Interface). Let $[\![\cdot]\!]$ be a realization functor from a cohesive or cubical HoTT universe to Euclidean parameter manifolds, mapping type-theoretic identity paths $p : \mathcal{O}_A =_U \mathcal{O}_B$ to continuous curves $[\![p]\!] : [0,1] \to \mathbb{R}^D$. If the realized curve $[\![\mathrm{ua}(e)]\!]$ coincides with the linear interpolation $\theta(t) = (1-t)\theta_A + t\,\sigma_{\pi^*}(\theta_B)$, then by Proposition A.5 the empirical risk along this path satisfies

$$\max_{t\in[0,1]} L(\theta(t)) \le \max\bigl(L(\theta_A), L(\sigma_{\pi^*}(\theta_B))\bigr) + \tfrac{\lambda}{8}\varepsilon^2,$$

where $\varepsilon = \|\sigma_{\pi^*}(\theta_B) - \theta_A\|$ and λ bounds $\nabla^2 L$ along the segment. This statement is a bookkeeping bridge from HoTT to Proposition A.5, not an independent proof that univalence produces the line segment.

Conjecture B.2 (Univalent LMC Interface)**.** *There exists a useful cohesive semantics for neural-network orbit packages in which: (i) permutation actions are represented by HIT quotient paths; (ii) alignment algorithms return proof objects consisting of orbit witnesses plus endpoint-distance certificates; (iii) univalence composes these certificates as paths between orbit packages; and (iv) realization maps such paths to Euclidean curves whose loss is controlled by second-order quantities such as those in Proposition A.5. The motivation is that each component already exists separately—group-action invariance, LSAP endpoint-distance minimization for matched blocks, and smoothness-based barrier bounds—but a single proof-relevant semantics combining them remains open.*

B.6. Alignment Certificates and a Conditional Theorem

We close with a concrete certificate statement that can be read independently of any proof-assistant implementation.

Definition B.3 (Alignment Certificate)**.** Let $\theta_A, \theta_B : \mathcal{N}$ be trained parameters and let G be the permutation-symmetry group acting by $\sigma : G \times \mathcal{N} \to \mathcal{N}$. An ε-*alignment certificate* from θ_B toward θ_A is a tuple

$$\mathsf{Cert}_\varepsilon(\theta_A, \theta_B) = (\pi,\ p_\pi,\ d_\pi),$$

where $\pi \in G$, $p_\pi : [\theta_B] =_{\mathcal{N}//G} [\sigma_\pi(\theta_B)]$ is the HIT quotient path generated by the group action, and

$$d_\pi : \|\theta_A - \sigma_\pi(\theta_B)\| \le \varepsilon$$

is a numerical endpoint-distance witness. For architectures with stateful normalization, the certificate may be extended by a calibration witness c_π specifying how interpolation-time statistics are recomputed.

Assumption B.4 (Linear Realization of a Certificate)**.** Given $\mathsf{Cert}_\varepsilon(\theta_A, \theta_B) = (\pi, p_\pi, d_\pi)$, the Euclidean realization used for the linear LMC test is

$$\gamma_\pi(t) = (1-t)\theta_A + t\,\sigma_\pi(\theta_B), \qquad t \in [0,1].$$

The formal path p_π records the representative change inside the quotient; the curve γ_π is an additional analytic choice in parameter space.

Theorem B.5 (Certified Linear Barrier Bound)**.** *Let* $\mathsf{Cert}_\varepsilon(\theta_A, \theta_B) = (\pi, p_\pi, d_\pi)$ *be an alignment certificate. Suppose* $\mathcal{L}$ *is twice differentiable along* γ_π*,* $\nabla^2\mathcal{L} \preceq \lambda I$ *on this segment, and*

$$\nabla\mathcal{L}(\theta_A) = 0, \qquad \nabla\mathcal{L}(\sigma_\pi(\theta_B)) = 0.$$

Then the aligned linear-interpolation barrier satisfies

$$\max_{t\in[0,1]} \mathcal{L}(\gamma_\pi(t)) \le \max\bigl(\mathcal{L}(\theta_A), \mathcal{L}(\sigma_\pi(\theta_B))\bigr) + \frac{\lambda}{8}\varepsilon^2.$$

Proof. By the distance witness d_π, $\|\sigma_\pi(\theta_B) - \theta_A\| \le \varepsilon$. Applying Proposition A.5 to the segment from θ_A to $\sigma_\pi(\theta_B)$ gives

$$\max_{t\in[0,1]} \mathcal{L}(\gamma_\pi(t)) \le \max\bigl(\mathcal{L}(\theta_A), \mathcal{L}(\sigma_\pi(\theta_B))\bigr) + \frac{\lambda}{8}\|\sigma_\pi(\theta_B) - \theta_A\|^2.$$

Substituting the certificate bound yields the claim. □

Remark B.6 (Role of Univalence). The theorem above is an ordinary Euclidean statement once the certificate is fixed. HoTT contributes the proof-relevant quotient witness p_π and the ability, in a univalent universe of orbit packages, to compose and transport such witnesses. It does not by itself provide the numerical witness d_π, the Hessian bound, or the calibration witness for normalization layers; those remain empirical or analytic parts of the certificate.

B.7. Lean 4 Formulation of the Certificate Theorem

The following Lean 4 formulation mirrors Theorem B.5 at the level of the certificate interface. The quotient path is represented as an abstract field because ordinary Lean 4 does not natively provide the HIT quotient path used in HoTT; a cubical or HoTT kernel would replace this field by the corresponding path constructor. The analytic content is the theorem `certifiedLinearBarrier`: once the certificate and the second-order barrier lemma are available, the final bound follows by applying the distance witness.

```
import Mathlib.Analysis.Normed.Group.Basic
import Mathlib.Data.Real.Basic

/- Abstract Lean 4 interface for Appendix B.7.
   Param is the Euclidean parameter space and Sym is the
   permutation-symmetry group. -/

universe u v

variable {Param : Type u} {Sym : Type v}
variable [NormedAddCommGroup Param]

/-- An alignment certificate from thetaB toward thetaA. -/
structure AlignmentCertificate
    (act : Sym -> Param -> Param)
    (OrbitPath : Param -> Param -> Prop)
    (thetaA thetaB : Param) (eps : Real) where
  pi : Sym
  orbitWitness : OrbitPath thetaB (act pi thetaB)
  distWitness : norm (thetaA - act pi thetaB) <= eps

variable
  (loss : Param -> Real)
  (segment : Param -> Param -> Real -> Param)

/-- Euclidean second-order barrier lemma used as the analytic input. -/
axiom symmetricBarrierBound
    (thetaA thetaB' : Param) (eps lambda : Real) :
    norm (thetaA - thetaB') <= eps ->
    (forall t : Real, 0 <= t -> t <= 1 ->
      loss (segment thetaA thetaB' t)
        <= max (loss thetaA) (loss thetaB') + lambda / 8 * eps ^ 2)

/-- Certified linear barrier bound induced by an alignment certificate. -/
theorem certifiedLinearBarrier
    (act : Sym -> Param -> Param)
    (OrbitPath : Param -> Param -> Prop)
```

```
    (thetaA thetaB : Param) (eps lambda : Real)
    (cert : AlignmentCertificate act OrbitPath thetaA thetaB eps) :
    forall t : Real, 0 <= t -> t <= 1 ->
      loss (segment thetaA (act cert.pi thetaB) t)
        <= max (loss thetaA) (loss (act cert.pi thetaB))
             + lambda / 8 * eps ^ 2 := by
  exact symmetricBarrierBound loss segment
    thetaA (act cert.pi thetaB) eps lambda cert.distWitness
```